Supplementary Material

When Missing Days Change a Photovoltaic Comparison Decision: A No-Imputation Reliability Surface Across Three Field Datasets

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S1. Protocol freeze and notation

The R22 surface used in the manuscript was frozen before inspection of the combined result. The declared grid was

$$q \in \{0, 0.025, 0.05, 0.075, 0.10\}, \qquad L \in \{1, 2, 3, 4, 5, 6, 7\} \text{ whole days},$$

with at least ten retained days. Deleted days were not reconstructed. A source-month is counted as fragile at (q, L) if at least one contiguous gap of length no greater than L changes the candidate set relative to the full admitted month. The source-level surface is descriptive and months are not assumed independent.

The main-text shortlist is

$$S_m(q; D) = \{j : Y_{m,j}(D) \geq (1 - q) \max_k Y_{m,k}(D)\}.$$

For a contiguous gap $G_m(s, \ell)$, the month-level indicator is

$$F_m(q, L) = \mathbf{1}[\exists\, \ell \leq L, s : S_m(q; D_m \setminus G_m(s, \ell)) \neq S_m(q; D_m)].$$

No generic mathematical novelty is claimed for case-removal sensitivity or minimum deletion budgets.

S2. Dataset-specific admission details

S2.1 PROPRE.MA

The manuscript uses the exact 25 complete, boundary-stable source-months retained at the R18 gate. These months descend from source-locked cumulative-counter and date-identity work and preserve site/configuration-specific STC denominators. The cohort is not the entire 20-site historical dataset and does not re-admit historically excluded sites merely because a numerical record exists. Raw donor files are not redistributed. The source decision object is a three-technology normalized-yield shortlist.

S2.2 EURAC

The public archive is EURAC Research, Zenodo DOI 10.5281/zenodo.18316996. The source archive checksum used in R19 matched the published MD5. Four systems with defensible numeric total STC values were admitted: Trina poly-Si (2070 W), SunPower mono-Si (1200 W), Sanyo HJT (3870 W), and Wurth CIGS (1050 W). First Solar was excluded from the primary normalized analysis because a defensible total system STC could not be reconstructed from supplied metadata. Daily DC energy was based on common 15-minute P_{mpp} timestamps and a 95% common-irradiance support rule. Seven months passed the full-month gate. The supplied metadata do not fully establish identical mounting geometry, so results are described as observed-system decisions rather than intrinsic technology efficiency comparisons.

S2.3 MORE-EU

The public source is MORE-EU/OpenData, file `Solar Park Data -INACCESS/Array Power.csv`, pinned Git blob `5611277efdd581bb839ec380239846a43ebe0ba7`. The source reports normalized output for 14 arrays. A calendar day is admitted only when it contains exactly 96 rows and 96 unique quarter-hour labels. A month is complete only when every source-calendar day passes that rule. Seventy-nine months are admitted. Source-clock days with 92 or 100 rows are excluded rather than rescaled or shifted. Arrays are not assigned technology labels.

S3. Full decision-reliability surfaces

Tables S1–S3 report the cumulative fraction of admitted months for which at least one contiguous gap of length $\leq L$ changes the shortlist.

Table 1: *
Table S1. PROPRE.MA surface (25 admitted months).

q	$L \leq 1$	$L \leq 2$	$L \leq 3$	$L \leq 4$	$L \leq 5$	$L \leq 6$	$L \leq 7$
0%	0%	0%	0%	0%	0%	0%	0%
2.5%	0%	4%	4%	4%	4%	8%	12%
5%	4%	8%	12%	12%	12%	16%	16%
7.5%	12%	12%	16%	16%	16%	16%	20%
10%	4%	4%	4%	4%	8%	16%	16%

Table 2: *
Table S2. EURAC surface (7 admitted months).

q	$L \leq 1$	$L \leq 2$	$L \leq 3$	$L \leq 4$	$L \leq 5$	$L \leq 6$	$L \leq 7$
0%	0%	0%	0%	0%	14.3%	14.3%	14.3%
2.5%	0%	0%	0%	0%	0%	0%	0%
5%	14.3%	14.3%	14.3%	14.3%	14.3%	14.3%	14.3%
7.5%	0%	0%	0%	0%	0%	0%	0%
10%	0%	0%	0%	0%	0%	0%	0%

Table 3: *
Table S3. MORE-EU surface (79 admitted months).

q	$L \leq 1$	$L \leq 2$	$L \leq 3$	$L \leq 4$	$L \leq 5$	$L \leq 6$	$L \leq 7$
0%	6.3%	7.6%	8.9%	11.4%	11.4%	11.4%	11.4%
2.5%	51.9%	64.6%	69.6%	73.4%	73.4%	77.2%	82.3%
5%	44.3%	51.9%	57.0%	62.0%	62.0%	63.3%	64.6%
7.5%	17.7%	29.1%	31.6%	35.4%	35.4%	39.2%	39.2%
10%	11.4%	12.7%	15.2%	16.5%	19.0%	19.0%	20.3%

S4. Locked $q = 5\%$ cross-stage verification

The R22 expansion was required to reproduce the preceding R21 structured-gap result at $q = 5\%$ and $L \leq 3$ days. The cross-check is exact:

Dataset	R21 locked count	R22 count	Status
PROPRE.MA	3/25	3/25	PASS
EURAC	1/7	1/7	PASS
MORE-EU	45/79	45/79	PASS

The MORE-EU surface was independently recomputed from the pinned public Git blob and reproduced the same 79 complete source-calendar months. The upstream EURAC and PROPRE.MA cohorts were not redefined for R22.

S5. Interpretation and claim boundaries

The following statements are retained:

- the discrete candidate-set decision can be stable or fragile under whole-day gaps;
- contiguous gap length and declared decision margin must be reported together;
- dependence on q is non-monotonic in all three datasets at seven-day support;
- a missing-rate threshold alone is not a certificate of shortlist stability for this decision object;
- the three datasets provide source-diverse qualitative convergence without supporting a pooled probability.

The following statements are not supported by the present evidence:

- universal probability that a real PV outage changes an engineering decision;
- intrinsic technology superiority or procurement recommendation;
- causal climate explanation for a shortlist change;
- interpretation of q as measurement accuracy or uncertainty;

- statistical independence of months within a source;
- novelty of case-removal fragility, deletion radius, or minimum deletion budgets.

S6. Relation to prior work

The manuscript deliberately distinguishes its applied decision object from established missing-data and robustness literatures. Koubli et al. infer missing PV monitoring data to recover performance metrics; Livera et al. develop data-quality filtering and reconstruction; IEA PVPS provides technical-KPI guidance; Quest et al. assess missing gaps in performance-loss estimation; and Wang and Zhang develop mask-only policies for repairing incomplete PV monitoring records. Outside PV, minimal removal and case-influence ideas are well established in decision reliability, AMIP, training-sample removal, and ranking stability. The present work therefore positions the contribution as a PV-domain audit of a declared set-valued downstream decision rather than a new generic robustness theorem.

S7. Reproducibility and rights boundary

Public reproducibility materials are available at GitHub: pv-missing-day-decision-reliability. The repository releases derived surfaces, R22 claim/result locks, analysis and verification code, and manuscript/Supplement source; automated checks rebuild the surface figures and review PDFs. It does not redistribute PROPRE.MA donor raw data or third-party EURAC/MORE-EU raw data. EURAC and MORE-EU replay sources are identified by DOI or pinned Git blob, whereas PROPRE.MA raw-source replay requires authorized donor-archive access.

The R25 public-reproducibility integration changes no upstream R18–R22 numerical result. A future journal-specific version should not change the scientific surface unless a verified error or newly admitted evidence requires a new research stage.